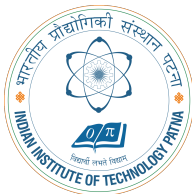


Chapter 05: BSS, ESS & Network Topologies

Dr. Mayank Agarwal

Department of CSE
IIT Patna

CS6206 Selected Topics in Wireless Networks



Lecture 5 Learning Objectives I

By the end of this lecture, you will be able to:

- Differentiate between IBSS, BSS, and ESS network architectures
- Explain the role of the Distribution System (DS) in enterprise Wi-Fi
- Describe the complete roaming process and its challenges
- Understand mesh networking concepts (802.11s)
- Design appropriate Wi-Fi topologies for different scenarios
- Calculate channel reuse patterns for multi-AP deployments
- Analyze trade-offs between different network topologies

The 802.11 Network Building Blocks I

- **Station (STA):** Any device with 802.11 capability
 - Client devices: laptops, phones, IoT devices
 - Can also be APs in some modes
- **Access Point (AP):** Bridge between wireless and wired networks
 - Coordinates access to the medium
 - Authenticates and associates stations
 - Often has wired Ethernet connection
- **Basic Service Set (BSS):** Fundamental building block
- **Extended Service Set (ESS):** Multiple BSSs appearing as one network

Key Concept: Wi-Fi networks are built by combining these building blocks in different configurations.

Independent BSS (IBSS): The Ad-Hoc Network I

Independent BSS (IBSS): Ad-hoc Network, no AP, devices talk directly to each other

- **Also known as:** Ad-hoc mode, peer-to-peer Wi-Fi
- **Characteristics:**
 - No central coordinator (AP)
 - Stations communicate directly with each other
 - Self-organizing, temporary networks
- **BSSID:** Randomly generated 48-bit address starting with 0x02
- **Beacon Generation:** Stations take turns sending beacons
- **No DS:** No connection to wired network

Example Use Cases:

- File sharing between laptops in a meeting
- Multiplayer gaming without internet
- Emergency communications when infrastructure is down

Limitations: Poor scalability, no internet access, security challenges

Infrastructure BSS: The Common Model I

Has an Access Point (AP) All communication goes through AP

- **Characteristic:** Centralized coordination via AP
- **BSSID:** MAC address of AP's radio interface
- **SSID:** Human-readable network name (32 bytes max)
- **Communication Pattern:** All traffic goes through AP
 - STA→STA communication: STA1 → AP → STA2
 - Enables power saving (AP buffers for sleeping STAs)
 - Simplifies management and security

AP Functions:

- 1 Beacon transmission and timing synchronization
- 2 Authentication and association of STAs
- 3 Frame buffering for power-saving STAs
- 4 Bridging between wireless and wired networks
- 5 Often implements security (WPA2/3, firewalls)

Numerical Example 1: BSS Capacity Analysis I

Problem: An 802.11ac AP supports maximum PHY rate of 1.3 Gbps. In practice, efficiency is about 50% due to overhead. The AP serves:

- 10 STAs doing web browsing (avg 100 Kbps each)
 - 5 STAs streaming video (2 Mbps each)
 - 2 STAs doing large file transfers (saturate available bandwidth)
- ❶ What's the practical maximum throughput of the BSS?
 - ❷ Is the AP the bottleneck for web browsing STAs?
 - ❸ What happens when file transfers are active?

Solution Steps:

- ❶ Practical throughput: $1.3 \text{ Gbps} \times 50\% = 650 \text{ Mbps}$
- ❷ Web browsing demand: $10 \times 100 \text{ Kbps} = 1 \text{ Mbps}$ (negligible)
- ❸ Video streaming demand: $5 \times 2 \text{ Mbps} = 10 \text{ Mbps}$

Numerical Example 1: BSS Capacity Analysis II

- ④ File transfers: Will compete for remaining $650 - 11 = 639$ Mbps
- ⑤ With DCF fairness: Each STA gets roughly equal airtime, not equal bandwidth

Model Answer:

- ① Practical BSS capacity: 650 Mbps
- ② AP not bottleneck for browsing/video: Only $1 + 10 = 11$ Mbps of 650 used
- ③ With file transfers: Rate anomaly occurs - low-rate STAs get equal airtime but less bandwidth. If 2 file transfer STAs and 15 other STAs share equally, each gets $\sim 1/17$ of airtime. File transfers get ~ 38 Mbps each, others get proportional to their rates.

Extended Service Set (ESS): Seamless Mobility I

Seamless roaming (move between APs without disconnecting)

- **Definition:** Multiple APs with same SSID connected via DS
- **Appearance:** Single logical network to STAs
- **BSSIDs:** Different for each AP (MAC addresses)
- **SSID:** Same across all APs in ESS

Requirements for ESS:

- 1 Same security settings (WPA2/3 passphrase or 802.1X)
- 2 Same VLAN configuration (for IP subnet consistency)
- 3 Fast roaming support (802.11r, OKC, PMK caching)
- 4 Coordinated channel planning

Enterprise Example: Campus Wi-Fi with hundreds of APs, one "University-WiFi" SSID. Example, "IITP-WiFi".

Distribution System (DS): The Backbone I

Think of the Distribution System (DS) as the invisible wiring that connects all Access Points (APs) inside an ESS

- **Purpose:** Interconnects APs within an ESS
- **Typical Implementation:** Switched Ethernet (Layer 2)
- **Alternative:** Wireless DS (mesh, WDS)
- **Key Functions:** If you move from AP1 → AP2, DS ensures your data still reaches you
 - Forward frames between APs for roaming
 - Distribute authentication state
 - Carry management traffic to controller (if used)

DS Components:

- **Portal:** Gateway to other networks (typically router)
- **Distribution System Medium (DSM):** Physical medium (Ethernet)
- **Distribution System Services (DSS):** Logical services (frame forwarding)

Architecture Choice: Autonomous APs vs. Controller-based

Autonomous vs. Controller-Based Architectures I

Portal → Gateway to internet (router)

DSM (Medium) → Physical link (Ethernet cable)

DSS (Services) → Logical work (forwarding, routing inside ESS)

Aspect	Autonomous (Fat AP)	Controller-Based (Thin AP)
Intelligence	Each AP independent	APs dumb, controller smart
Configuration	Per-AP configuration	Centralized configuration
Roaming	Layer 2 only, slow	Fast roaming (Layer 2/3)
Scalability	Poor (10s of APs)	Excellent (1000s of APs)
Cost	Lower for small deployments	Higher initial, lower Total Cost of Ownership large
Failure Impact	Single AP failure	Controller failure catastrophic
Examples	Home routers, small business	Enterprise (Cisco, Aruba)

Modern Trend: Cloud-managed APs (best of both?)

Roaming Process: How STAs Move Between APs I

This explains how the phone/devices switches from one AP to another.

- ❶ **Decision:** STA monitors current AP signal quality
 - Triggers: Low RSSI, high retry rate, missed beacons
 - Thresholds configurable (RSSI < -75 dBm typical)
- ❷ **Scanning:** STA scans for alternative APs
 - Background scan while connected
 - Active: Send probe requests
 - Passive: Listen for beacons

Chooses best AP based on:

 - Signal strength
 - Network load
 - Channel usage
- ❸ **Evaluation:** STA compares candidate APs
 - Metrics: RSSI, channel utilization, client count
- ❹ **Reassociation:** STA associates with new AP
- ❺ **Context Transfer:** Security, QoS state moved to new AP

Challenge: Seamless handoff with minimal packet loss

Roaming Latency Breakdown I

- **Discovery Time:** 50-200 ms (scan channels)
- **Authentication:** 5-50 ms (Open System fast, 802.1X slower)
- **Reassociation:** 5-10 ms
- **802.1X Re-authentication:** 100-500 ms (if required)
- **4-Way Handshake:** 20-50 ms (WPA2)
- **IP Renewal:** 0-1000+ ms (if new subnet)

Total: 80 ms to 2+ seconds

Voice/Video Requirements: <50 ms handoff for seamless experience

Solutions:

- **802.11r (FT):** Fast Transition (pre-authentication)
- **PMK Caching:** Reuse previous authentication
- **OKC:** Opportunistic Key Caching (proprietary)
- **802.11k:** Neighbor reports reduce scan time

Numerical Example 2: Roaming Threshold Calculation I

Problem: A STA moves away from AP1. Path loss exponent $n = 3.5$. At distance $d_1 = 20$ m, $RSSI = -55$ dBm. STA should roam before RSSI drops below -75 dBm for VoIP.

- 1 At what distance will RSSI reach -75 dBm?
- 2 If moving at **1.5 m/s**, how much time until roaming is needed?
- 3 How many VoIP packets (20 ms interval) may be missed during a 100 ms roam?

Formula used:

RSSI stands for Received Signal Strength Indicator. It is a measurement of how strong a wireless signal is when it reaches your device. Used in Wi-Fi, Bluetooth, and other wireless systems.

$$RSSI_2 = RSSI_1 - 10n \log_{10} \left(\frac{d_2}{d_1} \right)$$

Substitute given values: Usually measured in dBm (decibels relative to 1 milliwatt)

$$RSSI_1 = -55, \quad RSSI_2 = -75, \quad n = 3.5, \quad d_1 = 20, \quad d_2 = ?$$

Numerical Example 2: Roaming Threshold Calculation II

$$-75 = -55 - 10(3.5) \log_{10} \left(\frac{d_2}{20} \right)$$

$$-75 = -55 - 35 \log_{10} \left(\frac{d_2}{20} \right)$$

Rearrange:

$$-75 + 55 = -35 \log_{10} \left(\frac{d_2}{20} \right)$$

$$-20 = -35 \log_{10} \left(\frac{d_2}{20} \right)$$

$$20 = 35 \log_{10} \left(\frac{d_2}{20} \right)$$

Numerical Example 2: Roaming Threshold Calculation III

$$\log_{10} \left(\frac{d_2}{20} \right) = \frac{20}{35} = 0.571$$

$$\frac{d_2}{20} = 10^{0.571} = 3.72$$

$$d_2 = 20 \times 3.72 = 74.4 \text{ m}$$

Distance where RSSI reaches -75 dBm:

74.4 m

Time until roam is needed:

$$\Delta d = 74.4 - 20 = 54.4 \text{ m}$$

Numerical Example 2: Roaming Threshold Calculation IV

$$t = \frac{\text{distance}}{\text{speed}} = \frac{54.4}{1.5} = 36.27 \text{ s}$$

$$t \approx 36.3 \text{ s}$$

VoIP packets missed during 100 ms roam:

$$\text{Packets} = \frac{100 \text{ ms}}{20 \text{ ms}} = 5$$

5 packets

Final Answer:

- 1 Roam threshold is reached at **74.4 m**
- 2 Time until roaming is needed is **36.3 s**
- 3 Up to **5 VoIP packets** may be missed during roaming

Layer 2 vs. Layer 3 Roaming I

- **Layer 2 Roaming:** STA moves within same IP subnet
 - APs on same VLAN Layer 2 Roaming (Same Subnet):
Device moves between APs within the same VLAN / subnet
 - STA keeps same IP address IP address stays the same
 - Simpler, faster
 - Limited by broadcast domain size (<1000 devices)
- **Layer 3 Roaming:** STA moves to different IP subnet
 - APs on different VLANs/subnets
 - STA gets new IP address DHCP assigns new IP
 - Requires Mobile IP or similar Existing connections break (e.g., downloads, calls)
 - Complex, breaks existing TCP connections

Enterprise Design: It's complex : needs techniques like: Mobile IP, Tunneling, Session preservation mechanisms

- **Traditional:** Large Layer 2 domains for roaming
- **Modern:** Small Layer 3 domains with fast roaming (802.11r)
- **Controller-based:** APs tunnel traffic to controller (pseudo-Layer 2)

Mesh Networking (802.11s) I

- **Purpose:** Create wireless backbone without wired infrastructure
- **Standard:** 802.11s amendment (incorporated into 802.11-2012)
- **Key Features:** *A wireless backbone where APs connect to each other wirelessly instead of using cables.*
 - Self-forming, self-healing networks
 - Multi-hop routing (HWMP protocol)
 - Security (SAE, MSA)

Components:

- **Mesh Station (STA):** Participates in mesh
- **Mesh Gate:** Connects to external network
- **Mesh Path Selection:** HWMP (Hybrid Wireless Mesh Protocol)

Use Cases:

- Community networks
- Temporary deployments (events, disaster recovery)
- Extending coverage where wiring is difficult

Mesh vs. Traditional Infrastructure I

Traditional Wi-Fi

AP → Ethernet → Network

Mesh Wi-Fi

AP → AP → AP (wireless hops)

Aspect	Traditional Infrastructure	Mesh (802.11s)
Backhaul	Wired Ethernet	Wireless
Deployment Speed	Slow (cabling)	Fast (place nodes)
Cost	High (cabling labor)	Lower (no cables)
Reliability	Single point (AP)	Multi-path
Capacity	High (dedicated backhaul)	Lower (shared wireless)
Latency	Low	Higher (multiple hops)
Scalability	Good	Limited (wireless hops)

Rule of Thumb: Mesh useful where wiring is impossible/expensive, but performance degrades with each hop.

Mesh is convenient but slower. Best when wiring is impractical

Wireless Distribution System (WDS) I

Older method to connect APs wirelessly (before mesh standard)

- **Purpose:** Connect APs wirelessly (before 802.11s)
- **Operation:** APs act as bridges and clients simultaneously
- **Limitations:**
 - Vendor proprietary implementations
 - Typically halves throughput (receive then forward)
 - No standardized multi-hop routing
- **Common Use:** Home Wi-Fi extenders

AP acts as:

Client (receives data)
Bridge (forwards data)

WDS vs. Mesh:

- WDS: Simple point-to-point or star
- Mesh: Self-organizing, multi-hop, standardized

Channel Planning for Multi-AP Deployments I

Avoid interference between APs

- **Goal:** Minimize co-channel interference
- **2.4 GHz:** 3 non-overlapping 20 MHz channels – 1, 6, 11
- **5 GHz:** typically 24 non-overlapping 20 MHz channels if you count the full unlicensed 5 GHz range, including DFS channels; that becomes about 12×40 MHz, 6×80 MHz, or 2×160 MHz channels.
- **6 GHz:** in the full US 6 GHz allocation (5925–7125 MHz), there are 59×20 MHz channels, which group into 29×40 MHz, 14×80 MHz, 7×160 MHz, or 3×320 MHz channels.

Planning Approaches:

- 1 **Reuse Pattern:** Like cellular networks
- 2 **Power Control:** Reduce power to shrink cell size
- 3 **Channel Width:** Narrower channels = more non-overlapping

Channel Planning for Multi-AP Deployments II

Example 2.4 GHz Pattern:

1	6	11
6	11	1
11	1	6

Rule: No same-channel APs should be adjacent

Numerical Example 3: Channel Planning Calculation I

Problem: Design Wi-Fi for a $60\text{ m} \times 40\text{ m}$ office floor. AP coverage radius $\approx 30\text{ m}$. Using 5 GHz with 40 MHz channels.

- 1 How many APs are needed for coverage?
- 2 How many if using 80 MHz channels?
- 3 What is a good channel plan for 40 MHz channels in 5 GHz, assuming 9 non-overlapping 40 MHz channels are available?

Step 1: Floor area

$$A_{\text{office}} = 60 \times 40 = 2400\text{ m}^2$$

Step 2: Ideal AP coverage as a circle

$$A_{\text{circle}} = \pi r^2 = \pi(30)^2 = 2827\text{ m}^2$$

Numerical Example 3: Channel Planning Calculation II

This is the *ideal* coverage area, but circles do not pack efficiently without overlap or gaps.

Step 3: Practical planning with hexagonal cells

In wireless planning, each AP cell is often approximated by a **regular hexagon** because hexagons tile the floor efficiently.

The area of a regular hexagon is:

$$A_{\text{hex}} = \frac{3\sqrt{3}}{2}a^2$$

For a regular hexagon, the side length a is taken approximately equal to the AP range r , so:

$$A_{\text{hex}} = \frac{3\sqrt{3}}{2}r^2 \approx 2.6r^2$$

Numerical Example 3: Channel Planning Calculation III

For $r = 30$ m:

$$A_{\text{hex}} \approx 2.6(30)^2 = 2.6(900) = 2340 \text{ m}^2$$

Why the Hexagon Formula is Used I

Derivation of hexagon area:

A regular hexagon can be divided into **6 equilateral triangles**.

$$A_{\triangle} = \frac{\sqrt{3}}{4}a^2$$

So total hexagon area is:

$$A_{\text{hex}} = 6 \times \frac{\sqrt{3}}{4}a^2$$

$$A_{\text{hex}} = \frac{3\sqrt{3}}{2}a^2$$

Since

$$\frac{3\sqrt{3}}{2} \approx 2.598 \approx 2.6$$

Why the Hexagon Formula is Used II

we use the shortcut:

$$A_{\text{hex}} \approx 2.6r^2$$

Interpretation:

- Circle area πr^2 gives ideal RF coverage.
- Hexagon area $2.6r^2$ gives a more realistic **packing model**.
- This helps estimate the number of APs needed on a floor.

Number of APs for this office:

$$N = \frac{A_{\text{office}}}{A_{\text{hex}}} = \frac{2400}{2340} \approx 1.03$$

Round up:

2 APs minimum

In practice, walls, furniture, and capacity needs may increase this to **3–4 APs**.

For 40 MHz channels:

- Assume **9 non-overlapping 40 MHz channels** are available in 5 GHz.
- Since only about **2 APs** are needed for basic coverage, channel assignment is easy.
- Example plan: AP1 → Channel 1, AP2 → Channel 2
- If 3–4 APs are installed for practical reasons, assign different channels to adjacent APs.

For 80 MHz channels:

- Fewer non-overlapping channels are available than with 40 MHz.
- Coverage area is mainly determined by propagation and obstacles, so AP count stays about the same for coverage.

Channel Planning Answer II

- However, channel reuse becomes harder because there are fewer distinct channels.

Model Answer:

- 1 Minimum APs for coverage:

2 APs

- 2 With 80 MHz channels:

AP count is still about 2 for coverage, but channel reuse is harder

- 3 Optimal 40 MHz channel plan:

Assign different non-overlapping channels to adjacent APs

With 9 available 40 MHz channels, this is straightforward.

Multi-Band Operation (Dual/Tri-Band) I

- **Dual-Band:** 2.4 GHz + 5 GHz
- **Tri-Band:** 2.4 GHz + 5 GHz + 5 GHz (or 6 GHz)
- **Benefits:**
 - Load balancing between bands
 - Legacy support on 2.4 GHz
 - High performance on 5/6 GHz
 - Dedicated backhaul in mesh systems
- **Band Steering:** Encourage 5 GHz adoption
 - Delay 2.4 GHz probe responses
 - Prefer 5 GHz in association
 - Can cause compatibility issues

Client Behavior: Modern devices support both bands, may prefer 5 GHz

Virtual APs and Multiple SSIDs I

- **Virtual AP (VAP):** Single physical AP presenting multiple BSSIDs
- **Typical Limit:** 8-16 SSIDs per radio
- **Use Cases:**
 - Separate networks: Corporate, Guest, IoT
 - Different security policies per SSID
 - Different VLANs per SSID
- **Overhead:** Each SSID adds beacon overhead
 - More airtime consumed
 - Clients must process all beacons

Example Configuration:

- SSID "Corp-WiFi": WPA2-Enterprise, VLAN 10
- SSID "Guest-WiFi": Captive portal, VLAN 20, rate limited
- SSID "IoT-WiFi": WPA2-Personal, VLAN 30, IoT devices

Limitation: All SSIDs share same channel and airtime

Virtual AP (VAP): Basic Idea I

Definition: A **Virtual AP (VAP)** is a **single physical AP** that presents **multiple logical wireless networks**.

- One physical AP can broadcast multiple **SSIDs**
- Each virtual network has its own **BSSID**
- To clients, these appear as separate Wi-Fi networks
- All are created by the same AP hardware and software

Example:

- OfficeWiFi
- GuestWiFi
- IoTWiFi

Key point: One real AP behaves like multiple APs logically.

SSID, BSSID, and VAP Relationship I

SSID: The network name seen by users

Example: OfficeWiFi

BSSID: The unique MAC address identifying one wireless network instance

VAP concept:

- Each VAP forms a separate **Basic Service Set (BSS)**
- Each BSS has its own **BSSID**
- A single physical AP can create multiple BSSIDs

Example:

- OfficeWiFi → BSSID 1
- GuestWiFi → BSSID 2
- IoTWiFi → BSSID 3

Thus: One physical AP → multiple VAPs → multiple BSSIDs

Why VAPs Are Used I

Benefits:

- Separate networks for employees, guests, and IoT devices
- Different security settings for each SSID
- Different VLAN mapping for traffic separation
- Better policy control using the same AP hardware

Important limitation:

- Multiple VAPs on one radio still share the same:
 - channel
 - airtime
 - radio capacity
- Too many SSIDs increase beacon overhead

Conclusion: VAPs improve flexibility and network segmentation, but they do **not** create extra radio capacity.

Load Balancing and Client Steering: Why Needed I

Problem: Clients usually connect to the AP with the **strongest signal**, not necessarily the **best AP**.

- A nearby AP may already be overloaded
- Another AP with slightly weaker RSSI may offer better performance
- This creates **uneven client distribution**

Resulting issues:

- Too many clients on one AP
- Reduced throughput per user
- Increased contention for airtime
- Poor user experience even when nearby APs are underutilized

Goal:

- Spread clients more evenly across available APs
- Improve overall network efficiency
- Reduce congestion on overloaded APs

Load Balancing and Client Steering Methods I

Load Balancing: Tries to distribute clients across APs before or during association.

- **Reject association requests** if an AP is overloaded
- **Suggest another AP** that has lower utilization
- Enforce **client limits per AP** to prevent overcrowding

Client Steering: Encourages already connected clients to move to a better AP.

- **Deauthentication-based steering:**
 - AP disconnects the client temporarily
 - Client reconnects, hopefully to a better AP
 - Effective but disruptive
- **Power adjustment:**
 - Reduce AP transmit power
 - Makes neighboring APs more attractive

- **Ignoring sticky clients:**
 - AP may ignore probes from clients that should roam away
 - Encourages reconnection to a different AP

802.11v: BSS Transition Management

- Allows an AP to **suggest** that a client move to another AP
- The AP provides information about candidate APs
- The **client makes the final decision**
- More graceful than forced steering methods

Why it is useful:

- Helps reduce load on congested APs
- Improves roaming efficiency
- Supports better client distribution in enterprise WLANs

Ethical / operational consideration:

- **Deauth-based steering** breaks connectivity temporarily
- This may interrupt ongoing traffic such as:

- VoIP calls
 - video streams
 - real-time applications
- Therefore, standards-based and less disruptive steering is preferred

High-Density Deployments I

Definition: A high-density WLAN is a deployment where many users share a limited wireless area, such as stadiums, lecture halls, airports, and convention centers.

Challenges:

- **Co-channel interference:** nearby APs on same channel interfere
- **Limited airtime:** many users compete for transmission opportunities
- **Many clients per AP:** overload reduces performance
- **Beacon overhead:** too many SSIDs/APs consume airtime

Solutions:

- Deploy **more APs at lower power** (small-cell design)
- Use **careful channel planning** to reduce interference
- Apply **band steering** to move clients to 5 GHz

High-Density Deployments II

- Use **airtime fairness** instead of throughput fairness
- Set **client limits per AP** to avoid overcrowding

Example: Stadium Wi-Fi

- Thousands of clients in a small area
- APs placed under seats, on aisles, or with directional antennas
- Aggressive steering and careful planning are essential
- 2.4 GHz is often disabled to reduce interference

Outdoor vs. Indoor Deployments I

Aspect	Indoor	Outdoor
Interference	Walls, multipath	Weather, foliage (trees, bushes, and plants)
Coverage Area	Small cells (20-30m)	Large cells (100m+)
Antennas	Omnidirectional	Directional/Panel
Channel Planning	Reuse patterns critical	Fewer APs, less reuse needed
Obstacles	Walls, furniture	Trees, buildings, terrain
Power	Low (20 dBm typical)	High (up to 30 dBm)
Regulations	Less restrictive	More restrictive (EIRP limits)

Topology Design Principles I

- ➊ **Coverage vs. Capacity:** More APs = higher capacity but more interference
- ➋ **Channel Reuse:** Minimize co-channel interference
- ➌ **Band Selection:** Use 5/6 GHz for capacity, 2.4 GHz for coverage
- ➍ **AP Placement:** Avoid corners, central locations better
- ➎ **Power Settings:** Lower power often better in dense deployments
- ➏ **Future Proofing:** Plan for more devices, higher bandwidth

Rule of Thumb:

- Office: 1 AP per 1000-1500 sq ft
- Classroom: 1 AP per classroom
- High-density: 1 AP per 15-25 users
- Outdoor: 1 AP covers 200-500 ft radius

Site Survey Process I

1 Pre-Survey:

- Floor plans, construction materials
- User density and requirements
- Existing infrastructure

2 On-Site:

- Test AP placements
- Measure signal strength
- Identify interference sources

3 Post-Survey:

- Create heat maps
- Finalize AP locations and channels
- Document installation plan

Tools: Ekahau, AirMagnet, Wi-Fi analyzers, spectrum analyzers

Wi-Fi Heat Maps in Deployment I

Definition: A **Wi-Fi heat map** is a color-coded visual representation of wireless coverage over a floor plan or deployment area.

What heat maps can show:

- **RSSI / Signal strength** across different locations
- **SNR** (signal-to-noise ratio)
- **Coverage gaps** or dead zones
- **AP overlap** and interference regions
- Expected **throughput / capacity** in different areas

How heat maps are created:

- **Predictive heat map:** Generated before deployment using floor plans, walls, AP locations, transmit power, and antenna details
- **Survey heat map:** Created after deployment by measuring real signal values on-site

Interpretation:

- Strong coverage areas may appear in one color
- Weak or unusable coverage areas appear in another color
- This makes wireless behavior easy to visualize

How Wi-Fi Heat Maps Help I

Benefits in Wi-Fi deployment:

- Identify **dead zones** and weak coverage areas
- Check whether AP placement provides **full floor coverage**
- Detect excessive **overlap** between neighboring APs
- Support better **channel planning** and power tuning
- Ensure enough overlap for **smooth roaming**
- Reveal the effect of **walls, glass, and obstacles**
- Help with **capacity planning** in high-density areas

Overall purpose:

- Place APs more effectively
- Reduce interference
- Improve user experience

- Validate network design after installation

Conclusion: Heat maps help convert Wi-Fi deployment from guesswork into a more accurate and optimized design process.

Common Topology Mistakes I

- 1 Too few APs: Coverage gaps, poor performance
- 2 Too many APs: Excessive interference, cost
- 3 Poor channel planning: Co-channel interference
- 4 Center of room placement: Actually suboptimal (corner-ish better)
- 5 Ignoring vertical propagation: Signals travel through floors
- 6 No capacity planning: Only considering coverage
- 7 Mixing vendors: May lack interoperability features
- 8 Ignoring future growth: Under-provisioning

- **Wi-Fi 6/6E:** OFDMA for better multi-user efficiency
- **Wi-Fi 7:** Multi-Link Operation (use multiple bands simultaneously)
- **Cloud-Managed:** Central control without on-prem controller
- **AI/ML Optimization:** Dynamic channel and power adjustment
- **Private 5G Integration:** Complementary technologies

Future Trend: Converged wireless (Wi-Fi + cellular + IoT) managed as one system

Summary: Key Topology Concepts I

- ❶ **BSS:** Single AP with associated STAs (basic cell)
- ❷ **ESS:** Multiple BSSs with same SSID (seamless network)
- ❸ **DS:** Backbone connecting APs (typically Ethernet)
- ❹ **Roaming:** STA movement between APs with minimal disruption
- ❺ **Mesh:** Wireless backhaul for flexible deployment
- ❻ **Channel Planning:** Critical for minimizing interference
- ❼ **Capacity vs. Coverage:** Fundamental design trade-off

- **Required Reading:**

- Textbook (Gast): Chapter 5 - "WLAN Topologies"
- Review IBSS vs BSS vs ESS differences

- **Optional Reading:**

- IEEE 802.11-2020: Clause 4.3 (BSS/ESS concepts)
- 802.11s amendment for mesh details

- **Next Lecture (Lecture 6): Network Planning & Management**

- Channel selection and interference management
- Site survey principles
- Capacity vs coverage tradeoffs
- Enterprise architectures

Review Questions I

- 1 What's the difference between BSSID and SSID?
- 2 Why does all traffic go through the AP in infrastructure mode?
- 3 Calculate: 10 APs on 2.4 GHz with only channels 1,6,11. Minimum reuse pattern?
- 4 Explain the trade-off between using 40 MHz vs 80 MHz channels.
- 5 What are the advantages and disadvantages of mesh networking?
- 6 Why might a client "stick" to a distant AP instead of roaming?
- 7 How does 802.11r improve roaming performance?
- 8 What problem do virtual APs (multiple SSIDs) solve?

Discussion Question: "You're designing Wi-Fi for a new 10-story office building. Would you use large Layer 2 domains for easy roaming, or smaller Layer 3 domains with fast roaming protocols? Consider management complexity, broadcast traffic, and failure domains."

Exercise: Network Topology Design

- **Objective:** Design a Wi-Fi network for a given scenario
- **Scenario:** 3-story school building, 50 classrooms, 1000 students
- **Tasks:**
 - 1 Determine number of APs needed
 - 2 Create channel plan for 2.4 GHz and 5 GHz
 - 3 Design ESS for seamless roaming
 - 4 Plan for multiple SSIDs (staff, students, guests)
 - 5 Consider capacity requirements (1:1 device program)
- **Deliverable:** Design document with diagrams and justifications

Learning Outcome: Apply topology concepts to real-world design problem.

① Assess Requirements:

- Coverage area (sq ft/meters)
- User density (users per area)
- Application needs (bandwidth, latency)
- Mobility requirements (roaming needed?)

② Choose Architecture:

- Single AP (home/small office)
- Multiple APs with wired backhaul (enterprise)
- Mesh (no wiring possible)

③ Determine AP Count:

- Coverage-based calculation
- Capacity-based calculation
- Use larger of the two

④ Plan Channels:

- Minimize co-channel interference
- Consider channel width trade-offs

5 Consider Special Areas:

- High-density (auditoriums, lobbies)
- Outdoor areas
- Secure areas (different VLANs/SSIDs)

Appendix: Roaming Algorithm Pseudocode I

```
def should_roam(current_ap, candidate_aps):  
    # Check current AP quality  
    if current_ap.rssi < ROAM_THRESHOLD:  
        # Find best candidate  
        best_ap = None  
        best_score = -inf  
  
        for ap in candidate_aps:  
            # Calculate score (higher is better)  
            score = calculate_ap_score(ap)  
  
            # Prefer 5 GHz over 2.4 GHz  
            if ap.band == "5GHz":  
                score += BAND_BONUS  
  
            # Prefer less loaded APs  
            load_penalty = ap.client_count * LOAD_FACTOR
```

Appendix: Roaming Algorithm Pseudocode II

```
score -= load_penalty

if score > best_score:
    best_score = score
    best_ap = ap

# Only roam if significantly better
if best_score > current_ap.score + HYSTERESIS:
    return best_ap

return None # Stay with current AP
```

Hop Count Impact on Throughput:

- Each wireless hop approximately halves throughput
- Reason: Same radio for receive and forward (half-duplex)
- With dedicated backhaul radio: Better but still degradation

Example: 100 Mbps connection to internet

- 1 hop: 50 Mbps to client (AP→Client)
- 2 hops: 25 Mbps (Internet→AP1→AP2→Client)
- 3 hops: 12.5 Mbps

Mesh Design Rule: Minimize hops, max 3-4 hops practical

Real-World Case Study: University Campus I

- **Scale:** 200 buildings, 5000 APs, 30,000 users
- **Architecture:** Controller-based with 50 controllers
- **Topology:**
 - Each building: Local Layer 3 domain
 - Fast roaming (802.11r) between buildings
 - Redundant controllers for reliability
- **SSIDs:**
 - Secure: WPA2-Enterprise (faculty/staff)
 - Student: WPA2-Enterprise with policy
 - Guest: Captive portal, rate limited
 - IoT: Separate VLAN for devices
- **Challenges:**
 - Interference between adjacent buildings
 - High density in lecture halls
 - Legacy device support

Thank you!